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Research

The career decision of information systems people *

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This study examines the relationships between career decisions and directions and a set of independent variables including job title, demographic variables, role stressors, boundary spanning activities, perceived job characteristics, and career outcomes for 348 IS employees. Results show that the majority had already defined specific jobs they would like to hold in the near future. They were also able to define four directions for their future careers: IS technical, IS management, business, and consultant. Results show that career decisions and directions are related to some of the independent variables. IS employees in each of the four career directions could be further categorized into groups on the basis of similar characteristics. Implications of theses findings are discussed.

Keywords: IS personnel; Career paths; Career decisions; Management of IS personnel

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1. Introduction

Lately both practitioners and researchers have shown increased concern over the career issues of information systems (IS) people. Two significant

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factors behind this interest are: (1) the widening gap between supply and demand of IS personnel; and (2) their reported high turnover rates. Considerable research has accumulated on: motivation [9,10], satisfaction and turnover [4,8,13,23], career planning [18], career pathing [11], job design [17,21], job performance [5,16], and the differences between IS and non-IS employees [12,25]. The focus of these studies and others is on the current and past career experience of IS employees, but little attention has been directed to examining the future career issues of these same IS employees.

Common to most technically oriented fields, decisions (e.g., such as whether to remain in the technical arena or move into management) eventually plague IS personnel and create internal conflict. Both the organization and the individual tend to suffer. Knowing opportunities in the current position, what jobs might be of future interest, suitability of the employee for these jobs, and the organization's willingness to work with the employee during a career planning process are pivotal to a successful human resource management system within the IS environment. Only limited research has been directed towards these issues. The purpose of this study is to better understand and document these processes.

Available IS literature on career pathing has been somewhat anecdotal and largely based on job histories. It typically recommends that organizations develop a dual career path for IS employees [7]. Having a career ladder for technical employees, and a separate but equal ladder for management was first recognized within the engineering environment in the mid 1950's. The concern at that point was that too many good engineers were opting for the more lucrative and prestigious management jobs. This often resulted in loosing technical resources and producing less-than-satisfactory managers. Transplanted technical personnel were inadequately trained as supervisors and soon became disenchanted.

Another emerging recommendation has been the creation of additional career paths which recognize the diversity of career desires and provides incentives for IS personnel to stay within a technical area. Such career ladders are wider and allow for horizontal movement at each rung. Laudon and Laudon [20] reported that many organizations have created new positions within user divisions called business systems analysts. Some organizations have retained the technical expertise but provide a prestige traditionally associated with managerial positions by creating senior-level in-house consultants.

The population of IS employees is quite diverse. Organizations must recognize that career opportunities, rewards, incentives, and organizational development programs currently available may be irrelevant or even inappropriate for different segments of this population. Organizational reward systems and career path opportunities must be flexible enough to accommodate a diverse workforce. If this occurs, employees will be able to make career decisions leading to more compatible job functions with higher levels of satisfaction. This should, in turn, lead to a more productive work force with reduced turnover.

Career decision was initially considered the ability of a person to choose a specific occupation [6]. Employees were either decided or undecided defending on their commitment to a specific occupational field. Generally, it was assumed that employees who are committed to a specific career path are more effective in career decision making and more successful in managing their careers. It follows that organizations able to facilitate career commitment of employees will directly benefit from it.

This study seeks to:

(1) determine if, for IS employees, there are relationships between specific career directions and demographic characteristics, career success outcomes, role stressors, job characteristics, boundary spanning activities and performance;
(2) examine the relationship between career decisions and job title; that is, the relationship between the employee's current job and the desired/future one;
(3) identify the degree to which career decisions are actively initiated by IS employees or are simply reactive responses to situations.

The study takes a more detailed look at career decisions, including career decision status and career direction, for a population that has already chosen to pursue the specific IS occupation. Career decision status refers to two categories of IS employees: Those who have and have not already selected career goals. The key issue here is whether they are able to develop career plans that move from the broad occupational level to a more specific set of decisions. For *career direction* we try to establish whether those with specific career goals have selected technical, management, business or consultant oriented roles.

The relationships of intent are:

Career decisions and demographic variables the relationships between career decisions and demographic variables. We hypothesize that as employees accumulate greater experience (age), organization (level), job (time in position), and occupation (years in the IS field), they can presumably assess themselves more accurately, gain insights into different career positions and work organizations, and know what jobs they would like to hold. We further propose that for IS personnel there is a positive relationship between level of education and career decisiveness.

Career decisions and levels of role conflict and ambiguity IS employees who are relatively career undecided should report greater levels of role ambiguity and role conflict. It is expected that people who have a relatively high tolerance for both role conflict and role ambiguity have less difficulty in processing career-related information and making career decisions, having greater confidence in career decision making. We further expect there to be relationships between the degree of interaction of IS personnel with end users and the levels of both role ambiguity and role conflict. There should be less role stress for IS employees who choose to be in technical positions and do not extensively deal with end users.

Career decisions and boundary spanning activities Boundary spanning activities require individuals to cross departmental or organizational boundaries in performing their job duties [3]. Recent research [1] indicates that the functions of system analysis and design require continual interaction with individuals, necessitating the crossing of many intra-organizational boundaries. It is expected that individuals intending to hold managerial or business jobs need to cross departmental and organizational boundaries more than technical employees.

Career decisions and job characteristics There is a high need for the freedom to be creative and original as well as to work on challenging tasks. Since IS employees need to have freedom to be creative and original, the lack of challenging tasks and autonomy may be related to career decisions. Other perceived job characteristics, as reflected by rewards, may also be related to current career and future job decisions. For example, IS employees who wish to hold technical positions may rely more heavily on the intangible rewards (freedom and autonomy, challenge, etc.). By examining the relationships between perceived job characteristics and career decisions we expect to find that employees who are assigned to challenging tasks and are able to pursue their ideas may be more comfortable making career decisions.

Career decisions and work attitudes We hypothesize that career decisions and work attitudes are related. IS employees who view their jobs or careers with a great deal of ambivalence are more likely to be undecided about their future careers. In addition, IS employees who are unsure whether to continue with their present organizations may alienate themselves from these organizations [6]. Since IS employees currently involved in supervision or considering shifts into management jobs perform a wide variety of work, they may be more satisfied with their jobs and careers, and be more committed to their organization. We expect that career decisions and future career jobs and salary levels are related.

2. Method

2.1. Sample and procedure

In early 1990, a sample copy of an *MIS Career Attitude Survey* was mailed to presidents of 20 Mid Atlantic chapters of the *Data Processing Management Association* (DPMA). Seven of these agreed to participate in this research project. Each chapter provided the authors with membership mailing lists, and a copy of the survey was mailed to each member. The survey was accompanied by a cover letter from the president of the chapter strongly encouraging participation. A postage-paid envelope was enclosed for return of completed surveys to the university address. Participation was strictly voluntary and members were assured that individual responses to the survey would be treated confidentially.

Surveys were mailed to 1152 DPMA members. Thirty one were returned because of incorrect addresses. Four hundred and fourteen completed surveys were received by the researchers for a 36.9% response rate. The elimination of surveys with missing data as well as surveys from respondents in non-MIS positions resulted in a final sample of 348 IS employees representing a broad cross-section of DPMA chapters in the Mid Atlantic Region. A summary of chapter representation and the demographic characteristics of the sample are presented in *Tables 1* and *2*.

2.2. Measures

Demographic items were included in the background information section of the survey. Participants indicated their job titles in an open-ended item. Age, organizational tenure, job tenure, and tenure in the DP/MIS field were all measured in years.

Career decision Respondents were asked whether they had decided what work position they would like to hold within the next 1 to 3 years period. Those who had not decided were coded 1 and the ones who already decided on a particular job were coded 2. In addition, those having further decided on a specific work function were classified into 4 different categories: Technical jobs (programmers), IS management, non-IS management (business), and consultants.

Role stressors Two types of role stressors were measured. *Role ambiguity* refers to the extent to which an employee is unclear about the job duties, performance standards, or level of performance. *Role conflict* refers to incompatible expectations of a work role. This often arises when two or more demands occur together.

Both role ambiguity and role conflict were measured with six and eight items respectively from scales developed by Rizzo et al. [24]. Each was scored on a 5-point scale ranging from (1) very false to (5) very true. The responses were coded so that the greater the score, the greater the perceived ambiguity or conflict. The six role ambiguity items (e.g., "I know exactly what is expected of me"; "I receive a clear explanation of what has to be done") were *averaged* to obtain an overall index of role ambiguity. The internal con-

Table 1
Membership represented in the sample.

Chapter name	*N* (members)	Percentage
Baltimore	130	37.4
Central Pennsylvania	49	14.1
Cumberland valley	15	4.3
Lehigh valley	29	8.3
Montgomery county	70	20.1
Pocono northeast	18	5.2
Schuylkill valley	37	10.6
Total	348	100.0

Table 2
Demographic characteristics of the sample in percentages of total ($N = 348$).

Gender	
Male (1) [a]	75.0
Female (2)	25.0
Job title	
Programmer (1)	14.3
Analyst (2)	9.8
Project leader (3)	7.9
MIS manager (4)	56.7
Other MIS (5)	11.3
Organizational level	
Professional staff or consultant (1)	37.3
First level supervisor (2)	10.8
Middle management (3)	38.8
Strategic management (4)	13.1
Education	
Some high school or less (1)	0.0
High school (2)	4.3
Some college (3)	27.3
Bachelor's degree (4)	23.0
Some graduate school (5)	17.1
Graduate degree (6)	28.3
Annual salary categories	
Below $25 000 (1)	4.9
25 000 – 34 999 (2)	10.7
35 000 – 44 999 (3)	21.0
45 000 – 54 999 (4)	26.2
55 000 – 64 999 (5)	15.6
65 000 or above (6)	21.6
Size of the computer department (No. of employees)	
1– 5 (1)	14.5
6–10 (2)	9.6
11–20 (3)	12.2
21–40 (4)	12.7
Above 40	51.0

[a] The numbers in parentheses represent the coding for the value of the specific variable.

sistency reliability (alpha) of the scale was 0.82. Similarly, the eight role conflict items (e.g., "I receive incompatible requests from two or more people;" "I have to do things that should be done differently") were also *averaged* to create an overall role conflict score (alpha = 0.87).

Perceived job characteristics Thirteen items were used to assess the extent to which an employee's current job provided opportunities and rewards. The response options were anchored on a five-point scale ranging from (1) not at all to (5) to a great extent. Eight items were adapted from Allen and Katz [2], and five were added to measure different aspects of the job. A factor analysis of the thirteen items produced two factors which explained almost 60% of the total variance.

Factor 1, *tangible rewards*, included: Work on projects leading to advancement, receive substantial annual salary increases, and receive a promotion within the next year or two. Responses to these three items were averaged to create a scale tapping the tangible component of the job (alpha = 0.79).

Factor 2, *intangible rewards*, included: Pursue your ideas, build a professional reputation, work with competent colleagues, work on technically challenging tasks, work on professionally important projects, have freedom to be creative and original, be highly respected by your peers, work on projects important to the organization, be highly respected by top management, and have a great deal of power and influence on the job. Responses to the ten items were averaged to create a scale representing the intangible component of the job (alpha = 0.90).

Boundary spanning activities A three-item scale was developed to assess boundary spanning activities: The degree to which employees interact and communicate with others outside their department. Each item (e.g., "assist other units in determining appropriate uses of information technology"; "recommend new applications of information technology to top management") was scored on a five-point scale ranging from (1) not part of my job to (5) a very significant part of my job. The three items were averaged to create the boundary spanning activities score (alpha = 0.89).

Job involvement This refers to the degree to which an employee identifies psychologically with the present job, and the extent to which the job situation is central to the employee's self-identity. Job involvement was measured by an eight-item scale adapted from Kanungo [19]. Individuals indicated their agreement or disagreement with each statement (e.g., "I consider my job to be very central to my existence"; "most of my interests are centered around my job") on a five point Likert-type scale ranging from (1) strongly disagree to (5) strongly agree. Responses to the eight items were averaged to create a job involvement score (alpha = 0.85).

Perceived performance IS employees responded to: "How would you rate your current performance?" Responses were made on a five-point scale with the following anchors: Unsatisfactory; marginal; good; excellent; and outstanding.

Career satisfaction This refers to the emotional reactions of employees to the progress they have achieved in their career. It was measured by a five-item scale adapted from prior research [14], with appropriate changes to make the items more relevant to the present sample. Individuals indicated their agreement or disagreement (e.g., "I am satisfied with the success I have achieved in my career"; "I am satisfied with the progress I have made toward achieving my overall career goals") on a five point Likert-type scale ranging from (1) strongly disagree to (5) strongly agree. Responses were averaged to create a career satisfaction score (alpha = 0.90).

Job satisfaction Job satisfaction refers to the emotional reactions of individuals to their jobs and their job experience. It was assessed with a three-item scale [15] reflecting overall satisfaction with the job. Each item (e.g., "Generally speaking I am satisfied with my job") required the respondents to indicate their feelings on a five point scale ranging from (1) strongly disagree to (5) strongly agree. Responses were averaged to produce a total job satisfaction score (alpha = 0.70).

Organizational commitment This variable, defined as the identification with a particular organization and the desire to maintain membership in that organization, was measured by an abbreviated version of the organizational commitment questionnaire (OCQ) developed by Porter et al.

[22]. The shorter version used in this study represents a more 'pure' measure of the affective dimensions of commitment to the employing organization. Sample items are "I really care about the fate of my organization"; and "I feel very little loyalty to my organization". The response options ranged from (1) strongly disagree to (5) strongly agree. They were coded so that high scores reflected greater commitment to the organization. The coefficient of reliability of this measure was 0.88.

3. Results and discussion

3.1. Career decisions

Results of the study show that 266 IS employees (76.4%) indicated they had a particular job position they would like to hold within the coming 1 to 3 years period. Only 82 (23.6%) of the IS employees were undecided.

3.2 Career direction

For those having decided about their future career directions, our data show that 23 (8.6%) indicated that they would like to hold technical positions (programmers), 33 (12.4%) would like to be consultants, 26 (9.8%) preferred non-IS (e.g., other business functional areas). The majority of IS employees, 184 (69.2%), indicated a desire to hold IS management positions. This indicates that the majority of IS employees prefer to stay in the IS field either holding technical or managerial positions (77.8%). Interestingly, few would like to cross to other business areas.

Results indicate that careers primarily fall into one of four categories. Two are within the IS area (technical and managerial), one within the more general organizational area (non-IS), and lastly, one outside the organization (consultant). Many of the IS employees would also like to be considered for management positions in such functional areas. It is apparent that organizations need to place greater emphasis on dual career paths within the IS department. Those options would provide greater incentives to IS employees to stay with their current organizations instead of seeking opportunities elsewhere.

3.3. Career decisions and career direction vs. job title

Table 3 shows the cross-tabulation of the career directions and job titles. Results show that there is no relationship between them. For those decisive IS employees who have indicated a specific job goal, Table 3 presents the frequency that these people desire to remain in each of the four career directions. As expected, a majority of people holding a consulting job would like to stay within the area (59%). It is also important to note that some employees currently in IS management would like to move to senior management positions outside the department. This re-inforces the earlier suggestion that organizations should not restrict IS employees from non-IS jobs and should consider training and educating these managers in other business functions. This would encourage them to remain in the organization where they could still achieve their non-IS positions. The organization could identify the educational and training requirements for these jobs and determine the fit between the employee and the job position. Based on previous findings, if there is a match between the employee job desire and the job title, there will be: More motivation, productivity, and satisfaction with the job and career; as well as loyalty to the current organization; and eventually less likelihood to leave.

3.4. Career decisions and career directions vs. demographic characteristics

Gender and career decisions are unrelated. About three quarters of the IS male and female

Table 3
The relationships between career directions and job title. [a,b]

Job title	Career directions			
	IS		Business	Consultant
	Technical	Management		
Programmers	20	23	8	9
Analysts	0	25	1	2
Project leaders	1	16	0	1
IS managers	2	113	15	8
Consultants	0	7	2	13

[a] The relationship between career directions and job title was found to be statistically significant at level 0.001.
[b] The values presented in this table are the number of people in a job title who would like to hold different jobs.

employees had made decisions regarding their future careers. Among these, more females wanted to hold technical positions than males (15.2% vs. 6.5%), but fewer females would like to be consultants (4.5% vs. 11.5%). These results may be a reflection of the reported difficulty that females encounter in seeking senior management positions or becoming entrepreneurs. The demographic data on gender also suggests a general under-representation of females. While they represent 45% of the employed US labor force, they only account for approximately 25% of positions in this study.

Table 4 indicates that age, job tenure and number of years in the IS field are unrelated to career decisions, but education and organizational tenure are related. Highly educated employees and those with fewer years of tenure with the organization decided what jobs they would like to hold more frequently than their counterparts.

Furthermore, *Table 5* shows that age, education, job tenure and number of years in the IS field are also related to decisions regarding career directions. Results indicate that those employees preferring to hold technical positions are typically older and have longer tenure in their jobs than those desiring other positions. This may be due to their lack of adequate management and business skills required for non-technical positions. It is interesting to note that the more highly educated employees would like to be consultants and the less educated ones would like to move to technical positions. This confirms the idea that highly educated employees typically see more options. Corporate human resource managers often refer to a perceived relationship between mobility of employees and educational level. Many organizations find that relocation of valuable employees away from their home area generates reluctance to accept offers from the less educated workers. Furthermore, a high percentage of those who accept the move eventually either leave the company or request transfers back to the original location. The physical and intellectual exposure at the college level is clearly more broadening than high school, and certainly there is a relationship between resistance to change and fear of the unknown.

Table 4
The relationships between career decisions and the study variables.

Independent variables	Career decisions	Career indecision	F-test [a]
Demographic variables			
Age	41.59	41.12	0.17
Education	4.52	4.09	8.60 ***
Organizational tenure	8.66	10.60	3.99 *
Job tenure	4.41	4.65	0.19
Number of years in the IS	15.87	14.59	1.33
Role stressors			
Role ambiguity	2.38	2.57	4.21 *
Role conflict	2.86	2.93	0.39
Boundary spanning activities	3.25	3.09	0.99
Perceived job characteristics			
Tangible rewards	3.07	2.79	3.71 *
Intangible rewards	3.92	3.72	5.50 *
Career outcomes			
Job satisfaction	3.77	3.59	3.95 *
Career satisfaction	3.81	3.68	1.49
Organizational commitment	3.79	3.58	6.05 *
Job involvement	2.84	2.56	3.55 *
Salary	4.07	3.67	3.82 *
Perceived performance	3.95	3.83	7.20 **

[a] * $p \leq 0.05$; ** $p \leq 0.01$; *** $p \leq 0.001$.

3.5. Career decisions and career directions vs. role stressors and boundary spanning activities

Role ambiguity is related to career decisions but role conflict is unrelated. IS employees who experience low levels of role ambiguity tend to know what job they would like to hold. IS employees who would like to move into the business category experience lower levels of ambiguity than others. These employees may consider their jobs and performance objectives to be clearly defined by their supervisors. Both the technical and managerial IS employees experienced similar low levels of role ambiguity.

On the other hand, IS employees who would like to be consultants reported having experienced higher levels of ambiguity. This may be due to the nature of their jobs. In consultant type positions, people frequently present a problem that is not precisely defined and consequently may serve as a source of ambiguity. Also, expectations from consultants are often intentionally vague – "study what you like as long as you help us". It is not unusual for them to be presented with the management's estimate of the problem, only to find that entirely different issues are at work. Furthermore, since their work involves a great deal of creativity and autonomy, there is additional opportunity for role ambiguity.

Boundary spanning activities are unrelated to career decisions. Conversely, among the IS employees who had decided what type of jobs they would like to hold, boundary spanning activities and career directions are related. IS employees who would like to hold IS management positions or business positions are engaged in more extensive boundary crossing than others. Their jobs require them to have continuous interaction with individuals both internal and external to the IS department. These employees are more likely to come in contact with individuals from a variety of functions and organizational levels. However, IS employees who would like to hold technical positions feel there is little need to cross departmental or organizational boundaries in order to perform their duties.

3.6. Career decisions and career directions vs. perceived job characteristics

Both tangible and intangible rewards are related to career decisions. Results indicate that employees who had decided what jobs they would like to hold perceive that these jobs would provide them with both tangible and intangible rewards. They evidently believe they know what those jobs offer and accordingly have decided on their course of action.

Intangible job rewards, such as autonomy and challenge, are related to career directions. IS employees desiring business jobs expressed the strongest relationship, followed by those seeking management and technical positions. IS employees who would like to be consultants were found to be the group least interested in intangible rewards. Possibly tangible rewards (money, advancement, and promotion) are more important to them. Conversely, those who desire technical

Table 5
The relationships between career directions and the study variables.

Independent variables	Career directions				F-test [a]
	IS		Business	Consultant	
	Technical	Management			
Demographic variables					
Age	47.33	40.07	43.14	41.37	4.53 ••
Education	4.00	4.37	4.52	5.28	5.04 ••
Organizational tenure	9.09	9.01	9.60	9.68	1.02
Job tenure	7.78	3.80	4.55	4.00	4.79 ••
Number of years in the IS	19.83	15.24	16.45	13.59	2.71 •
Role stressors					
Role ambiguity	2.39	2.39	2.10	2.62	2.52 •
Role conflict	2.86	2.89	2.66	2.93	0.57
Boundary spanning activities	2.53	3.34	3.33	3.18	2.83 •
Perceived job characteristics					
Tangible rewards	2.76	3.02	3.18	3.01	0.75
Intangible rewards	3.71	3.97	4.20	3.59	4.77 ••
Career outcomes					
Job satisfaction	3.77	3.76	4.07	3.52	2.78 •
Career satisfaction	3.73	3.84	4.00	3.49	1.71
Organizational commitment	3.73	3.77	3.89	3.77	0.24
Job involvement	2.73	2.78	2.55	2.66	0.86
Salary	2.83	4.17	3.57	4.56	8.82 •••
Perceived performance	3.87	3.95	3.96	4.04	0.31

[a] • $p \le 0.05$; •• $p \le 0.001$; ••• $p \le 0.001$.

positions may perceive that they had no chance for advancement. For those who see more advancement opportunities, moving up to either IS management or other functional areas is an appropriate direction. Note that, inconsistent with our expectations, there are no significant differences among the four career directions and their relationships to tangible rewards.

3.7 Career decisions and career directions vs. career outcomes

Result show that five out of the six independent career outcome variables are related to career decisions. Table 4 shows them to be: Job satisfaction; organizational commitment; job involvement; salary; and performance. Highly paid IS employees who are more satisfied and involved with their jobs, and more loyal to their organizations have a more decided attitude to their future careers. Those employees may have been receiving the feedback and information needed to help them in initiating future career plans. Surprisingly, a significant relationship was not found between career satisfaction and career decisions.

Since career outcomes and some of the other independent variables, such as role ambiguity, demographics, and perceived job characteristics, are related, we decided to control these predictors and test for differences in satisfaction, commitment, salary, involvement, and perceived performance. No significant differences were found, except in performance. This suggests that differences between the decisive and undecisive IS employees may be partially due to differences in variables of role ambiguity, demographic characteristics, and tangible and intangible rewards.

Those IS employees who are more satisfied with their jobs and careers would prefer career directions involving business (non-IS) jobs. On the other hand, consultants were found to be least satisfied with their jobs and careers. It is important to note that when controlling for role stressors, perceived job characteristics and demographics characteristics, these differences in both job and career satisfactions across the four career directions become non-significant. This suggests that the initial differences in job satisfaction and organizational commitment are due to differences in role stressors, perceived job characteristics and demographic characteristics.

It appears that IS employees earning relatively high salaries were most likely to be consultants, and least prefer an IS technical career direction. This perceived relationship between future career direction and salary level remained significant after controlling for role stressors, perceived job characteristics, and demographic characteristics. These independent variables do not appear to explain the differences in this relationships.

4. Summary and conclusion

This study has investigated the relationships between the independent variables of role stressors, boundary spanning activities, perceived job characteristics, career outcomes, and demographics, and the two dependent variables of career decisions, and future career directions of IS employees.

The major finding was that a majority of IS employees had already defined the specific jobs they would like to hold within the next one to three years. Furthermore, these decisive IS employees were able to define 4 directions in which they sought future careers: The IS technical area, IS management, non-IS area (business), and consultant areas. Whether we use these terms or the broader framework of: Working for an organization or working for themselves, is not the key issue. Of greater importance is the need for organizations to recognize that in order to retain and keep current IS personnel performing at high levels there need to be multiple career paths. There must be recognizable, attainable and relevant incentives and rewards. The danger lies in the possibility that many of these and future IS employees may not find a match between their career decision goals and directions, and the potential opportunities within the organization.

There are clear implications regarding human resource issues in managing IS employees. At a minimum, management should frequently monitor the progress of IS employees and provide them with the opportunity to advance within the organization. Concurrently, it is also important to incorporate dual career paths for those IS employees desiring to remain within the IS department. Career paths need to be designed with wider rungs on each step of the ladder. Organizations should encourage IS employees indicating an interest in moving up to senior management outside the IS department. This requires a proactive human resource intervention and not just a "go for it" attitude.

Although the DPMA members surveyed in this study represent a wide variety of organizations and functional areas, they were all members of one association and do not necessary represent the population of IS employees. Therefore, the transferability of these results to employees in other companies and other geographical areas is not proved. Finally, although bivariate analyses identified a number of relationships between career directions and decisions and job satisfaction, organizational commitment and intention to stay, and other variables, the observed relationships may be more complex than the present findings indicate.

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